Table II. Size distributions of eroded sediment

Experiment	Time period (minutes)	Sampling location S1 %Sa	S1 %Si	S1 %Cl	S2 %Sa	S2 %Si	S2 %Cl	CF1 %Sa	CF1 %Si	CF1 %Cl
1	0–4	55	24	21	60	22	18	30	35	35
	5–9									
2	0–4	47	32	21	43	38	19	68	19	13
	5–9	51	31	18				47	30	23
	10–14	43	38	19	40	40	20			
	15–19	42	39	19	40	41	19	42	34	24
3	0–4	50	29	21	44	32	24	73	12	15
	5–9									
4	0–4	49	30	21	83	7	10	49	26	25
	5–9				69	17	14			
	10–14	52	31	17	60	23	17	45	33	22
	15–19				59	23	18			
	20–24	66	19	15	60	26	14	52	25	23
	25–29									
Average over all experiments		50·6	30·3	19·1	55·8	26·9	17·3	62·2	26·7	15·9

Accordingly, individual samples were pooled to yield composite samples of sediment passing through S1, S2, and CF1 for each five-minute period during each of the four experiments.

Before particle size analysis could be undertaken, an additional consideration was the required weight of sediment. We chose to use an Andreasan pipette to determine the percentage of sand, silt, and clay in the samples. Combining the requirements of this technique with a concern to have a sample large enough to give a reliable estimate of percentages of sand, silt, and clay, we deemed a minimum of 10 g to be necessary. Even after pooling the individual samples as described, some of the composite samples weighed far less than 10 g. In these cases, samples from contiguous 5-minute periods within experiments were again pooled. In this way, with the exception of CF1 during E1 and minutes 5–9 of E2 for which the samples weighed only 8·0 and 9·1 g, respectively, samples used in the analysis were all in excess of 10 g. The results of the particle size analyses on these samples are presented in Table II.

ANALYSES OF SEDIMENT SIZE CHARACTERISTICS

The sediment collected at the miniature flumes at S1 and S2 and that passing through CF1 is, on average, finer than the matrix soil (Tables I and II). t-tests on the percentages of sand contained within the samples confirm that these percentages would not be expected in samples coming from the same population as the four samples of matrix soil ($p = 0{\cdot}007$, $p = 0{\cdot}01$, and $p = 0{\cdot}004$ for S1, S2, and CF1, respectively).

Sediment transported by interrill overland flow has been shown to be largely, if not entirely, supplied by raindrop detachment (e.g. Ellison, 1945; Young and Wiersma, 1973). Raindrop-detached sediment can be divided into that which is simply dislodged by the impact of the falling raindrops and that which is carried in splash-crown droplets. That the interrill sediment is finer than the matrix soil implies, therefore, that either there is selective detachment by raindrops or selective transport by the interrill flow. We have limited data obtained from experiments on the plot under rainfall of 80 mm hr^{-1} which show sediment transported in splash droplets to consist of 86 per cent sand, 14 per cent silt, and 0 per cent clay. These data suggest, therefore, that the fineness of the sediment in interrill flow relative to the matrix soil (Table I) is not due to selective detachment of finer particles by raindrops. Indeed, the data imply that transported sediment is finer